
Prueba de prácticas

Asignatura:

Fecha:

Nombre:

DNI:

Grupo de prácticas:

Modelo de examen:

Grupo de teoría (A o B):

Ejercicio 1

Determinar si la siguiente forma argumentativa es válida o inválida

$$p \rightarrow (q \wedge r), \quad \neg q \vee \neg r \quad \vdash \quad \neg p$$

En caso de ser inválida, indicar un contraejemplo.

```
prem1[{p_, q_, r_}] := (p ==> (q & r))  
prem2[{p_, q_, r_}] := ((!q) || (!r))  
concl[{p_, q_, r_}] := (!p)
```

```

n = 3;
FormaE[v_] := (prem1[v] ∧ prem2[v]) ⇒ concl[v];
tautologia = True;
p = Table[False, {t, n}];
For[i = 0, i < 2^n, i++,
  j = i;
  For[f = n, f > 0, f--,
    resto = Mod[j, 2];
    j = Floor[j / 2];
    If[resto == 0, p[[f]] = True, p[[f]] = False];
  ];
  If[FormaE[p], Null, tautologia = False];
];
If[tautologia, Print["La forma enunciativa es una tautología"],
  Print["La forma enunciativa no es una tautología"]]

```

La forma enunciativa es una tautología

Como la forma enunciativa es una ~~tautología~~ entonces es una argumentación válida

Ejercicio 2

En el conjunto D de los divisores positivos de 72, definimos la relación de divisibilidad $R = \{(x, y) \in D \times D \mid x \text{ divide a } y\}$.

```

Deje2 = Divisors[72]
CARTESIANO[conjuntos_] := Module[{i, j, k, cartesianotemp},
  cartesiano = Table[{conjuntos[[1, i]], {i, Length[conjuntos[[1]]]}];
  Do[cartesianotemp = {};
    Do[Do[AppendTo[cartesianotemp,
      Append[cartesiano[[k]], conjuntos[[i, j]]];,
      {k, Length[cartesiano]}];, {j, Length[conjuntos[[i]]]}];
    cartesiano = cartesianotemp; {i, 2, Length[conjuntos]}];
  cartesiano];

{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72}

```

```
cart = CARTESIANO[{Deje2, Deje2}]
```

```
{ {1, 1}, {2, 1}, {3, 1}, {4, 1}, {6, 1}, {8, 1}, {9, 1}, {12, 1}, {18, 1},
  {24, 1}, {36, 1}, {72, 1}, {1, 2}, {2, 2}, {3, 2}, {4, 2}, {6, 2},
  {8, 2}, {9, 2}, {12, 2}, {18, 2}, {24, 2}, {36, 2}, {72, 2}, {1, 3},
  {2, 3}, {3, 3}, {4, 3}, {6, 3}, {8, 3}, {9, 3}, {12, 3}, {18, 3},
  {24, 3}, {36, 3}, {72, 3}, {1, 4}, {2, 4}, {3, 4}, {4, 4}, {6, 4},
  {8, 4}, {9, 4}, {12, 4}, {18, 4}, {24, 4}, {36, 4}, {72, 4}, {1, 6},
  {2, 6}, {3, 6}, {4, 6}, {6, 6}, {8, 6}, {9, 6}, {12, 6}, {18, 6},
  {24, 6}, {36, 6}, {72, 6}, {1, 8}, {2, 8}, {3, 8}, {4, 8}, {6, 8},
  {8, 8}, {9, 8}, {12, 8}, {18, 8}, {24, 8}, {36, 8}, {72, 8}, {1, 9},
  {2, 9}, {3, 9}, {4, 9}, {6, 9}, {8, 9}, {9, 9}, {12, 9}, {18, 9},
  {24, 9}, {36, 9}, {72, 9}, {1, 12}, {2, 12}, {3, 12}, {4, 12},
  {6, 12}, {8, 12}, {9, 12}, {12, 12}, {18, 12}, {24, 12}, {36, 12},
  {72, 12}, {1, 18}, {2, 18}, {3, 18}, {4, 18}, {6, 18}, {8, 18},
  {9, 18}, {12, 18}, {18, 18}, {24, 18}, {36, 18}, {72, 18}, {1, 24},
  {2, 24}, {3, 24}, {4, 24}, {6, 24}, {8, 24}, {9, 24}, {12, 24},
  {18, 24}, {24, 24}, {36, 24}, {72, 24}, {1, 36}, {2, 36}, {3, 36},
  {4, 36}, {6, 36}, {8, 36}, {9, 36}, {12, 36}, {18, 36}, {24, 36},
  {36, 36}, {72, 36}, {1, 72}, {2, 72}, {3, 72}, {4, 72}, {6, 72},
  {8, 72}, {9, 72}, {12, 72}, {18, 72}, {24, 72}, {36, 72}, {72, 72} }
```

```
Reje2 = Select[cart, Mod[#[[2]], #[[1]]] == 0 &]
```

```
{ {1, 1}, {1, 2}, {2, 2}, {1, 3}, {3, 3}, {1, 4}, {2, 4}, {4, 4}, {1, 6},
  {2, 6}, {3, 6}, {6, 6}, {1, 8}, {2, 8}, {4, 8}, {8, 8}, {1, 9},
  {3, 9}, {9, 9}, {1, 12}, {2, 12}, {3, 12}, {4, 12}, {6, 12}, {12, 12},
  {1, 18}, {2, 18}, {3, 18}, {6, 18}, {9, 18}, {18, 18}, {1, 24},
  {2, 24}, {3, 24}, {4, 24}, {6, 24}, {8, 24}, {12, 24}, {24, 24},
  {1, 36}, {2, 36}, {3, 36}, {4, 36}, {6, 36}, {9, 36}, {12, 36},
  {18, 36}, {36, 36}, {1, 72}, {2, 72}, {3, 72}, {4, 72}, {6, 72},
  {8, 72}, {9, 72}, {12, 72}, {18, 72}, {24, 72}, {36, 72}, {72, 72} }
```

1. Determinar si R es una relación de orden utilizando el módulo de análisis de relaciones.

2. Dibujar su diagrama de Hasse.

```

AnalisisRB[A_, R_] := Module[{}, falloReflexiva = {};
  Do[If[MemberQ[R, {A[[n]], A[[n]]}], Null,
    AppendTo[falloReflexiva, A[[n]]], {n, Length[A]}];
  falloSimetrica = {};
  Do[If[MemberQ[R, {R[[m, 2]], R[[m, 1]]}], Null,
    AppendTo[falloSimetrica, {R[[m, 2]], R[[m, 1]]}], {m, Length[R]}];
  falloTransitiva = {};
  Do[Do[If[R[[p, 1]] == R[[q, 2]],
    If[MemberQ[R, {R[[q, 1]], R[[p, 2]]}], Null, AppendTo[
      falloTransitiva, {{R[[q, 1]], R[[q, 2]]}, {R[[p, 1]], R[[p, 2]]}}]]];,
    {q, Length[R]}];, {p, Length[R]}];
  falloAntisimetrica = {};
  Do[If[MemberQ[R, {R[[r, 2]], R[[r, 1]]]] &&
    ! (ToString[R[[r, 1]]] == ToString[R[[r, 2]]]), AppendTo[
      falloAntisimetrica, {R[[r, 1]], R[[r, 2]]}]]];, {r, Length[R]}];
  If[falloReflexiva == {}, Print["R es reflexiva"], Print[
    "R no es reflexiva, falla en los elementos: ", falloReflexiva]];
  If[falloSimetrica == {}, Print["R es simétrica"],
    Print["R no es simétrica, falla en los pares: ", falloSimetrica]];
  If[falloTransitiva == {}, Print["R es transitiva"], Print[
    "R no es transitiva, falla en los pares: ", falloTransitiva]];
  If[falloAntisimetrica == {}, Print["R es antisimétrica"],
    Print["R no es antisimétrica, falla en los pares: ",
      falloAntisimetrica]];
  If[Union[falloReflexiva, falloSimetrica, falloTransitiva] == {},
    Print["R es una relación de equivalencia"],
    Print["R no es relación de equivalencia"]];
  If[Union[falloReflexiva, falloAntisimetrica, falloTransitiva] == {},
    Print["R es una relación de orden"],
    Print["R no es relación de orden"]];];

```

AnalisisRB[Deje2, Reje2]

R es reflexiva

R no es simétrica, falla en los pares:

$\{ \{2, 1\}, \{3, 1\}, \{4, 1\}, \{4, 2\}, \{6, 1\}, \{6, 2\}, \{6, 3\},$
 $\{8, 1\}, \{8, 2\}, \{8, 4\}, \{9, 1\}, \{9, 3\}, \{12, 1\}, \{12, 2\},$
 $\{12, 3\}, \{12, 4\}, \{12, 6\}, \{18, 1\}, \{18, 2\}, \{18, 3\}, \{18, 6\},$
 $\{18, 9\}, \{24, 1\}, \{24, 2\}, \{24, 3\}, \{24, 4\}, \{24, 6\}, \{24, 8\},$
 $\{24, 12\}, \{36, 1\}, \{36, 2\}, \{36, 3\}, \{36, 4\}, \{36, 6\}, \{36, 9\},$
 $\{36, 12\}, \{36, 18\}, \{72, 1\}, \{72, 2\}, \{72, 3\}, \{72, 4\}, \{72, 6\},$
 $\{72, 8\}, \{72, 9\}, \{72, 12\}, \{72, 18\}, \{72, 24\}, \{72, 36\} \}$

R es transitiva

R es antisimétrica

R no es relación de equivalencia

R es una relación de orden

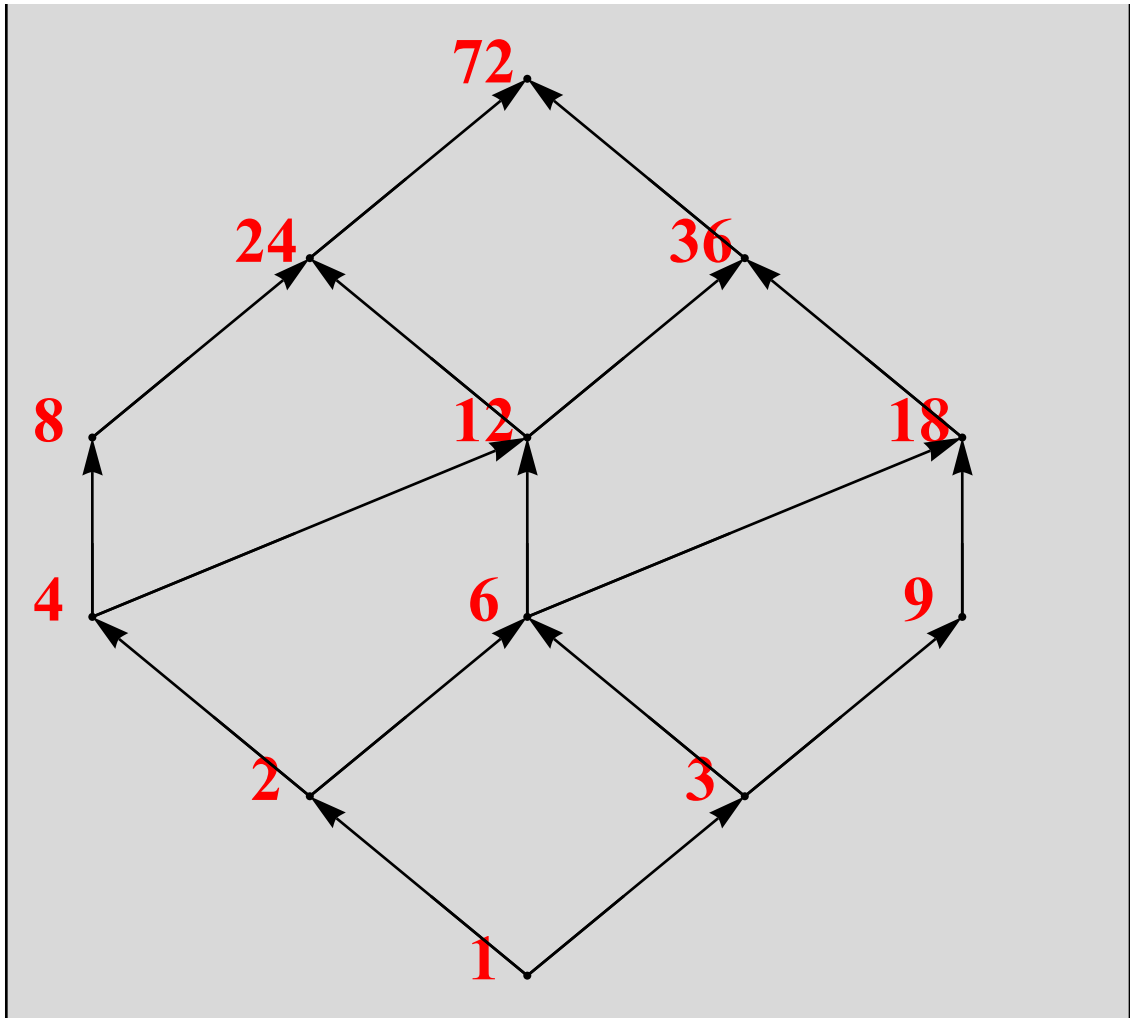
2. Dibujar su diagrama de Hasse.

```

A = Deje2;
R = Reje2;
Clear[Coord];
tabla = Table[0, {i1, Length[A]}, {j1, 3}];
B = A;
t1 = 1;
nivel = 0;
While[B ≠ {}, minimales = {}; nivel++;
  Do[minimal = True;
    Do[If[Intersection[{{B[[m1]], B[[n1]]}}, R] ≠ {} && n1 ≠ m1,
      minimal = False];, {m1, 1, Length[B]}];
    If[minimal, AppendTo[minimales, B[[n1]]];
      tabla[[t1]] = {nivel, B[[n1]], 0};
      t1++;];, {n1, 1, Length[B]}];
    B = Complement[B, minimales];];
R1 = {};
Do[AppendTo[R1, {A[[i1]], A[[i1]]}], {i1, 1, Length[A]}];
R = Complement[R, R1];
R1 = {};
Do[
  Do[If[Length[Intersection[R, {{R[[k1, 1]], A[[j1]]}, {A[[j1]], R[[k1, 2]]}}]] ==
    2, R1 = Union[R1, {R[[k1]]}];];,
    {j1, 1, Length[A]}, {k1, 1, Length[R]}];
R = Complement[R, R1];
puntos = {}; t1 = 0;
Do[cont = 0;
  Do[If[tabla[[i1, 1]] == j1, cont = cont + 1], {i1, 1, Length[A]}];
  Do[t1++;
    puntos = Union[puntos, {Text[Style[tabla[[t1, 2]],
      Large, Bold, Red, Background → None, FontFamily → Times],
      {k1 - (cont / 2) - .1, j1 + .1}], Point[{k1 - (cont / 2), j1}]}];
    tabla[[t1, 3]] = k1 - (cont / 2);, {k1, 1, cont}],
    {j1, 1, tabla[[Length[A], 1]}];
Coord[elem_] := Do[If[elem == tabla[[h1, 2]],
  Coord[elem] = {tabla[[h1, 3]], tabla[[h1, 1]]}, {h1, 1, Length[A]}];
Do[Coord[A[[i1]]], {i1, 1, Length[A]}];
Do[AppendTo[puntos, Arrow[{Coord[R[[t1, 1]]], Coord[R[[t1, 2]]]}];,
  {t1, 1, Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos], AspectRatio → 1, Background → None]

```

Diagrama de orden:



3. Sea $A = \{x \in D \mid x > 6 \text{ y } x \text{ es par}\}$. Calcular maximales, minimales, máximo y mínimo de A si existen.

```

MINIMO[A_, R_] := Module[{minimo, mini, n, m}, minimo = {};
  Do[mini = True;
    Do[
      If[MemberQ[R, {A[[n]], A[[m]]}], , mini = False;
      Break[]];, {m, 1, Length[A]}];
    If[mini, AppendTo[minimo, A[[n]]];
    Break[]];, {n, 1, Length[A]}];
  minimo];
  
```

```

MAXIMALES[A_, R_] := Module[{maximales, maximal, n, m}, maximales = {};
Do[maximal = True;
  Do[If[MemberQ[R, {A[[n]], A[[m]]}] && n ≠ m, maximal = False;
    Break[]];, {m, 1, Length[A]}];
  If[maximal, AppendTo[maximales, A[[n]]];, {n, 1, Length[A]}];
maximales]

```

```

MINIMALES[A_, R_] := Module[{minimales, minimal, n, m}, minimales = {};
Do[minimal = True;
  Do[If[MemberQ[R, {A[[m]], A[[n]]}] && n ≠ m, minimal = False;
    Break[]];, {m, 1, Length[A]}];
  If[minimal, AppendTo[minimales, A[[n]]];, {n, 1, Length[A]}];
minimales];

```

```

MAXIMO[A_, R_] := Module[{maximo, maxi, n, m}, maximo = {};
Do[maxi = True;
  Do[If[MemberQ[R, {A[[m]], A[[n]]}], , maxi = False;
    Break[]];, {m, 1, Length[A]}];
  If[maxi, AppendTo[maximo, A[[n]]];
    Break[]];, {n, 1, Length[A]}];
maximo]

```

```

A = {};

For[i = 1, i ≤ Length[Deje2], i++,
  If[Deje2[[i]] > 6 && EvenQ[Deje2[[i]]],
    AppendTo[A, Deje2[[i]]]
  ]
]

A

{8, 12, 18, 24, 36, 72}

```


MINIMALES [A, R]

MINIMO [A, R]

MAXIMALES [A, R]

MAXIMO [A, R]

{8, 12, 18}

{ }

{72}

{ }

Ejercicio 3

Considerando el conjunto D y el subconjunto A del ejercicio anterior:

```

RETICULO[A_, R_] :=
Module[{reticulo, cotassuperiores, cotasinferiores, csuper, cinfer,
  supremo, infimo, mini, maxi, n, m, x1, x2}, reticulo = True;
Do[Do[cotassuperiores = {}; cotasinferiores = {};
  Do[csuper = True; cinfer = True;
    If[Intersection[{A[[x1]], A[[n]]}], R] == {}, csuper = False];
    If[Intersection[{A[[x2]], A[[n]]}], R] == {}, csuper = False];
    If[Intersection[{A[[n]], A[[x1]]}], R] == {}, cinfer = False];
    If[Intersection[{A[[n]], A[[x2]]}], R] == {}, cinfer = False];
    If[csuper, AppendTo[cotassuperiores, A[[n]]];
    If[cinfer, AppendTo[cotasinferiores, A[[n]]];, {n, 1, Length[A]}];
  supremo = {}; infimo = {};
  Do[mini = True;
    Do[If[Intersection[{cotassuperiores[[n]], cotassuperiores[[m]]},
      R] == {}, mini = False], {m, 1, Length[cotassuperiores]}];
    If[mini, AppendTo[supremo, cotassuperiores[[n]]];,
    {n, 1, Length[cotassuperiores]}];
  If[supremo == {}, reticulo = False];
  Do[maxi = True;
    Do[If[Intersection[{cotasinferiores[[m]], cotasinferiores[[n]]},
      R] == {}, maxi = False], {m, 1, Length[cotasinferiores]}];
  If[maxi, AppendTo[infimo, cotasinferiores[[n]]];,
  {n, 1, Length[cotasinferiores]}];
  If[infimo == {}, reticulo = False];
  {x1, 1, Length[A]};
  {x2, 1, Length[A]};
reticulo
]

```

1. Razonar si D es un retículo.

RETICULO[Deje2, Reje2]

True

Es un retículo

2. ¿Es A un subretículo de D ? Justificar la respuesta.

SubsetQ[Deje2, A]

True

Como D es un retículo y A es un subconjunto no vacío de D entonces es un ~~subretículo~~ de D

3. Deducir si D con la estructura de retículo es un Álgebra de Boole (comprobar si es distributivo y complementario).

```
INFIMO[A_, B_, R_] :=
Module[{cotasinferiores, cinfer, infimo, maxi, n, m},
  cotasinferiores = {};
  Do[cinfer = True;
    Do[If[Intersection[{A[[n]], B[[m]]}, R] == {}, cinfer = False;
      Break[]];, {m, 1, Length[B]}];
    If[cinfer, AppendTo[cotasinferiores, A[[n]]];, {n, 1, Length[A]}];
  infimo = {};
  Do[maxi = True;
    Do[If[
      MemberQ[R, {cotasinferiores[[m]], cotasinferiores[[n]]}], , maxi = False;
      Break[]];, {m, 1, Length[cotasinferiores]}];
    If[maxi, AppendTo[infimo, cotasinferiores[[n]]];
      Break[]];, {n, 1, Length[cotasinferiores]}];
  infimo
]
```

```
RETICULODISTRIBUTIVO[A_, R_] :=
Module[{distributive, i, j, k}, distributivo = True;
  Do[
    Do[Do[If[TrueQ[SUPREMO[A, Union[{A[[i]]}, INFIMO[A, {A[[j]], A[[k]]}, R]],
      R] == INFIMO[A, Union[SUPREMO[A, {A[[i]], A[[j]]}, R], SUPREMO[A,
        {A[[i]], A[[k]]}, R]], R]], , distributivo = False];,
      {i, 1, Length[A]}];
    , {j, 1, Length[A]}];
    , {k, 1, Length[A]}];
  distributivo
]
```

```
COMPLEMENTADO[A_, R_] := Module[{complementado, k}, complementado = True;
  Do[If[COMPLEMENTO[A, R, A[[k]]] == {}, complementado = False],
    {k, 1, Length[A]}];
  complementado]
```

```
COMPLEMENTO[A_, R_, n_] := Module[{complemento, k}, complemento = {};
  Do[If[INFIMO[A, {A[[k]], n}, R] == MINIMALES[A, R] &&
    SUPREMO[A, {A[[k]], n}, R] == MAXIMALES[A, R],
    AppendTo[complemento, A[[k]]], {k, 1, Length[A]}];
  complemento]
```

MAXIMALES[Deje2, Reje2]

{72}

MINIMALES[Deje2, Reje2]

{1}

COMPLEMENTADO[Deje2, Reje2]

False

RETICULODISTRIBUTIVO[Deje2, Reje2]

True

Es un ~~alg~~ de Boole, ya que es un retículo distributivo y es un retículo con elemento 0 y 1, y no es complementado

Ejercicio 4

1. Calcular, si existe, el inverso de 23 en $\mathbb{Z}_{120}$.

```

Inverso2[x_, n_] :=
Module[{Cocientes, Euclides, Resto, a1, a2}, Euclides[n1_, n2_] :=
  If[Mod[n1, n2] == 0, n2, AppendTo[Cocientes, Quotient[n1, n2]];
    Euclides[n2, Mod[n1, n2]]];
Resto[1, x1_, x2_] := x1 - x2 * Cocientes[[1]];
Resto[2, x1_, x2_] := x2 - Resto[1, x1, x2] * Cocientes[[2]];
Resto[k_, x1_, x2_] :=
  Expand[Resto[k - 2, x1, x2] - Resto[k - 1, x1, x2] * Cocientes[[k]]];
a1 = Abs[x];
a2 = Abs[n];
Cocientes = {};
Euclides[a1, a2];
Return[Mod[Resto[Length[Cocientes], 1, 0] * Sign[x], n]]];];

```

Inverso2[120, 23]

14

2. Resolver el siguiente sistema de congruencias utilizando el Teorema Chino del Resto:

$$\begin{cases} 9x \equiv 2 \pmod{5} \\ 100x \equiv 3 \pmod{7} \\ 55x \equiv 4 \pmod{11} \end{cases}$$

Inverso2[5, 9]

2

Inverso2[7, 100]

43

Mod[43 * 3, 7]

3

Inverso2[11, 55]

1

```
AlgChino2[a_, m_] := Module[{M, b, Teorema, n, u, w, x0}, Teorema = "sí";
  n = Length[a];
  Do[Do[If[GCD[m[[i]], m[[j]]] ≠ 1, Teorema = "no";
    Break[]];, {j, i + 1, Length[m]}], {i, Length[m] - 1}];
  If[Teorema == "sí", Inverso[x_, n_] :=
    Module[{inverso}, Do[If[Mod[j * x, n] == 1, inverso = j;
      Break[]], {j, 1, n - 1}];
    Return[inverso]];
  M[1] = 1; M[i_] := M[i - 1] * m[[i - 1]];
  u[i_] := Inverso[M[i], m[[i]]];
  b[1] = Mod[a[[1]], m[[1]]];
  w[i_] := Mod[(a[[i]] - b[i - 1]) * u[i], m[[i]]];
  b[i_] := b[i - 1] + w[i] * M[i];
  x0 = b[Length[a]];
  Print["Solución: x = ", x0, " + t*", M[n + 1]];
  Print[TableForm[Join[{{a[[1]], m[[1]], 1, 1, "-", b[1]}}, Table[
    {a[[i]], m[[i]], M[i], u[i], w[i], b[i]}, {i, 2, n}]], TableHeadings →
    {Table[i, {i, n}], {"a", "m", "M", "u", "w", "b"}}]],];
  Print["No podemos aplicar el Tª Chino del Resto"];];];
```

AlgChino2[{4, 3, 4}, {5, 7, 11}]

Solución: $x = 59 + t \cdot 385$

	a	m	M	u	w	b
1	4	5	1	1	–	4
2	3	7	5	3	4	24
3	4	11	35	6	1	59